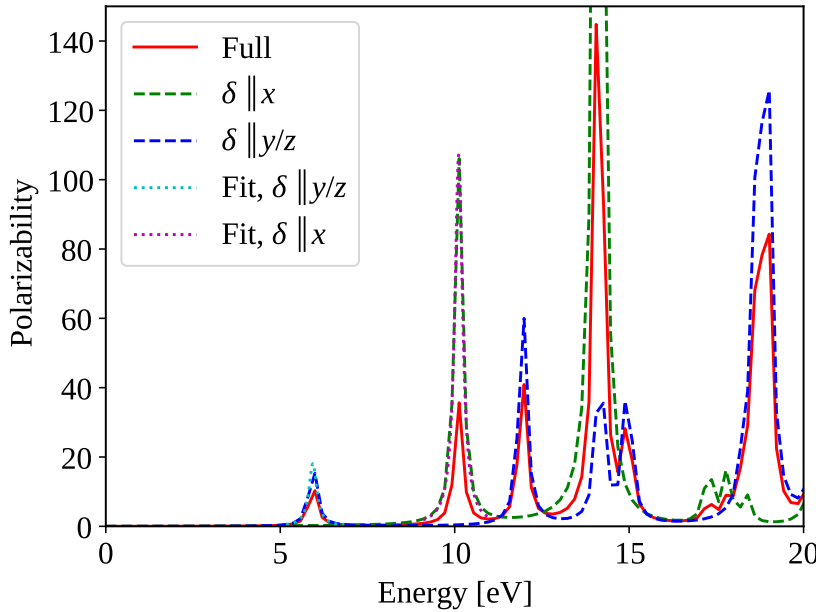


Determination of the pulse shape for CO molecule

Two low-lying excitations observed in broad spectra obtained from 1/

- excitation reaction to y pulse direction, equivalent with z direction, at energy $\varepsilon_y = 5.936$
- excitation reaction to x pulse direction, at energy $\varepsilon_x = 10.114$



Centre : 5.936 ± 0.001
Damping : 0.136 ± 0.002
Amplitude : 0.335 ± 0.006
Centre : 10.114 ± 0.002
Damping : 0.135 ± 0.001
Amplitude : 1.95 ± 0.04

In order to excite the oscillation selectively, apply pulse centered around the energy - need to determine the wavelength in nm.

$$E = hf = h \frac{c}{\lambda}$$

$$\lambda = \frac{hc}{E} = \frac{4.135 \text{ eV fs} \times 2.997 \times 10^2 \text{ nm/fs}}{E} \approx 1239 \frac{\text{eV nm}}{E}$$

So $\lambda_x \approx 122.529 \text{ nm}$, $\lambda_y \approx 208.770 \text{ nm}$.

In order to get a reasonably selective excitation, aim for pulse spread of cca 4 eV. From the Gaussian inversion in FT

$$\sigma_f = \frac{\sigma_E}{h} = \frac{1}{\sigma_t}$$

$$\sigma_t = \frac{h}{\sigma_E} \approx 1.00 \text{ fs}$$

Where I used $\sigma_E = 4.135 \text{ eV}$ to get a numerical cancelation.

It is reasonable to give the pulse time of $3\sigma_t$ before running into full force and we shall count the target time from this full force time, meaning that the propagation time should be 23 fs with pulse $T_0 = 3.00 \text{ fs}$. We should try varying the intensity until the moment amplitude in atomic units approaches reasonable fractions (percents) of unity.

TODO : Having some trouble with oscillating field, should implement this for debugging

- implement setting the start time for FTs in fs, look at previous code from eigenstate removal
 - maybe think about another section - move the keyword to RTBSE section?
- implement print section for field FT